CO2 - CO_2 Laser

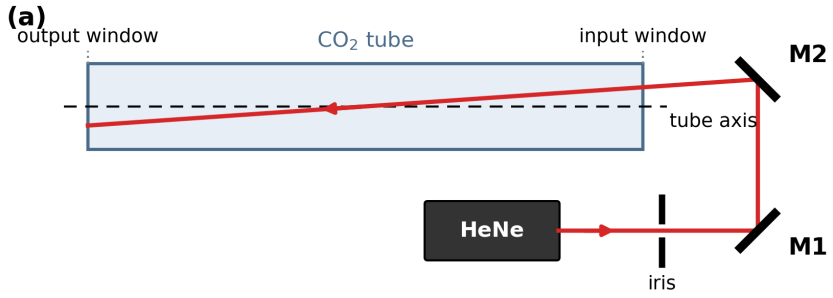
Signature Sheet

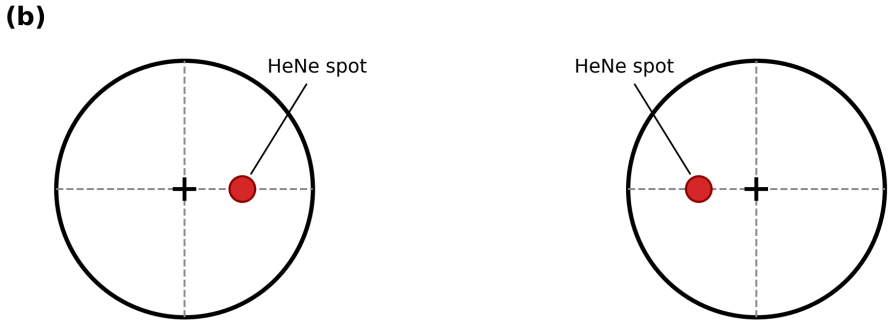
Student's Name _____ Partner's Name _____

Pre-Lab Discussion Questions

It is your responsibility to discuss this lab with an instructor before your first day of your scheduled lab period. This signed sheet must be included as the first page of your report. Without it you will lose grade points. You should be prepared to discuss at least the following before you come to lab:

1. How does a laser work, in general?
2. What is the purpose of the different gases (i.e. N_2, CO_2, and He)?
3. What are the necessary characteristics of the medium, and the cavity?
4. How are the practical requirements for laser action satisfied in a CO2 laser in this lab?
5. What does a grating do when one is used in place of a mirror in the laser?
6. To align this laser, you will steer the HeNe alignment beam down the center of the CO_2 tube bore using two mirrors, as sketched in figure below. Suppose you observe that the beam enters the tube to the *right* of center at the input window, and comes out to the *left* of center at the output window (both spots on the horizontal midline, see panel b). What is wrong with the beam — its position, its angle, or both? Can you fix it with the knobs of one mirror alone? Describe a procedure that brings the beam onto the tube axis. (Hint: a beam line in space is fixed by four numbers — two positions and two angles — and each mirror mount has only two knobs.)





beam position at the input window

beam position at the output window

7. What are the safety requirements for working with this laser?

Staff Signature _____ Date _____

Completed before the first day of lab? (Circle one) Yes / No

Mid-Lab Discussion Questions

1. On day 4 of this lab, you should have successfully aligned the laser and observed a lasing spectrum in the spectrum analyzer. How many lines are lasing? On what does this number depend? Demonstrate your progress to an instructor and ask for a signature.

Staff Signature _____ Date _____

Completed by day 4 of lab? (Circle one) Yes / No

Checkpoint Signatures

1. Laser Goggle Safety Check

   Staff Signature _____

2. Back Mirror Alignment

   Staff Signature _____

3. Proper Gas Flow

   Staff Signature _____

4. Spectrum Analyzer Signal Line

   Staff Signature _____

5. P(20) Line

   Staff Signature _____

6. CO_2 Spectrum

   Staff Signature _____

Station Clean-Up Check-Off

Before leaving, ensure your station is clean and ready for the next group. The GSI must verify and sign off on the following:

1. Any BNC cables that you have been moving are unplugged; return things to the state that you found them.
2. AFM prepared samples are removed and the sample disks are returned to the bag. CD/DVD/Blu-Ray scraps are thrown out if small, put in the bin if large.
3. All used pipette tips go into trash and glass sample slides go into glass disposal container; pipettes are put back to the shelf.
4. Workspace cleared of personal belongings, papers, and debris.
5. Any spills or hazards cleaned up; no left-over onion in the fridge or on the bench.
6. Microscope is clear of mounted sample, turned off and covered by the plastic bag properly if applicable.
7. Restart the computer at your station so that any software updates are applied.
8. Station is clean and ready for the next group. Staff discretion for additional cleanup that is necessary applies.

Staff Signature _____ Date _____